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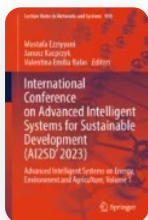
Artificial Intelligence and Employability: A Literature Review of Engineer's Competencies

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Abstract

Since the fourth Industrial Revolution, automation and machines have displaced human labor, and the advent of artificial intelligence (AI) is further transforming industries, necessitating engineers to adapt their competencies for employability. This literature review explores the intersection of AI and engineering, specifically focusing on the competencies essential for engineers in the AI-driven job market. AI's transformative impact on engineering, spanning design, manufacturing, maintenance, and data analysis, demands a reevaluation of conventional engineering skills. The review begins by scrutinizing the relevance of existing skills and the emergence of new AI-specific competencies.

As AI's role expands, engineers often lack expertise in areas like machine learning, data analytics, and algorithm development. Educational institutions are responding with innovative pedagogical models and curricular adjustments, yet the review stresses the ongoing need for education and lifelong learning to keep engineers abreast of this rapidly evolving field. Drawing from diverse research articles and resources, the study identifies specific engineering skills complemented by AI technologies.

Utilizing the Knowledge, Attitudes, Skills, and Habits (KASH) model, this paper comprehensively explores the evolving landscape of engineer competencies in the AI context. Emphasizing interdisciplinary collaboration between engineers and AI experts, the paper underscores the importance of acquiring AI-related skills and cultivating collaborative problem-solving habits. The methodology involves referencing the Scopus database for scientific publications on AI's utilization in engineering employment, contributing to a nuanced understanding of the evolving relationship between engineers and AI technologies.

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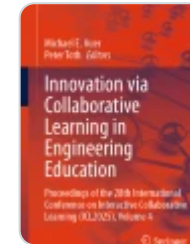
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